Rules for the Inspection, Maintenance and Repair of Locomotive Boilers

Standard rules governing the shop practice of the New York, New Haven and Hartford Railroad have appeared in the April and May issues. The rules continue as follows:

Tools for Working Tubes and Flues

All tools for working tubes or flues in locomotive boilers shall be railroad company's standard, and in accordance with railroad company's drawings where available.

There are three forms of sectional expanders:

1. Form F straight gaging.—Used only on flues 3½ inches and smaller for preliminary setting and tightening of flues.

2. Form B modified.—Used for tightening new flues before flues are beaded.

3. Form C straight finger.—Used for tightening ferrules, also for tightening flues on old work to eliminate leakage, etc.

All new sectional expanders shall be checked to standard contour gages. No expanders shall be used which do not conform to gages.

The tool room foreman is made responsible for checking the expanders as they are returned to the tool room and under no circumstances shall an expander with any appreciable wear again be issued from the tool room.

For the initial application of flues, 2-inch and 2¼-inch Faessler roller expanders with flaring rolls and arranged for both hand and power operation, shall be used.

For re-working flues in the front flue sheet, hand-operated expanders, or combined power and hand-operated expanders shall be used, size to be 2 inches or 2¼ inches as required.

Faessler roller expanders with flaring rolls and arranged for both hand and power operation, shall be used for 3-inch, 3½-inch, 4½-inch and 5½-inch flues.

Roller expanders are used at points making classified repairs and to a limited extent at engine houses.

Arch tube roller expanders of Faessler make are used in sizes 2⅞ inches, 3 inches, 3⅜ inches and 3½ inches. They are used only at points applying arch tubes.

Beading tools are supplied in two sizes, namely, for 2-inch, 2¼-inch and 3½-inch flues and for 5½-inch flues. No beading tools shall be used which do not conform to gages shown on company drawings. The tool room foreman is made responsible for checking the beading tools as they are turned in to the tool room, and under no circumstances shall a beading tool with any appreciable wear again be issued from the tool room.

All the above-mentioned tools shall be purchased and furnished on requisition.

Preparation of Boiler for Application of Tubes and Flues

On new back flue sheets, holes for 2-inch and 2¼-inch flues shall be the same diameter as the nominal outside diameter of flues.

Holes for the type "E" superheater flues shall be 3⅛ inches diameter. Holes for the type "A" superheater flues shall be 4⅝ inches diameter.

All holes in the back flue sheet for tubes and flues shall have the inside and outside edges rounded to a radius of 1/16 inch to prevent cutting of flues. When back flue sheet holes become ⅛ inch out of round, they shall be reamed. When, through service, 10 percent of the holes in the back flue sheet become enlarged 3/16 inch above the original diameter of the flue holes, the sheet shall be removed and scrapped.

In the front flue sheet, holes for 2-inch or 2¼-inch flues shall be 1/32 inch larger in diameter than the outside diameter of flues. Holes for type "A" flues shall be drilled 3/32 inch larger or 5 19/32 inches diameter. For type "E" flues, holes shall be drilled 3/32 inch large or 3 19/32 inches diameter.

All holes shall be cleaned and both the inside and outside edges rounded to a radius of 1/16 inch to prevent cutting of flues.

In the preparation of firebox door and flue sheets for arch tubes, the holes for arch tubes shall be drilled the same size as the outside diameter of arch tubes. The water side of sheets shall be rounded to a radius of 1/16 inch. On new sheets this shall be done while fabricating the sheet and before applying it to the boiler.

In applying 2-inch and 2¼-inch flues, clean the holes, if necessary, and place ferrules of proper thickness in the back flue sheet holes not more than 1/16 inch in from fire side. The utmost care shall be taken to observe this requirement as it is essential to proper welding of the flues. Tighten with form "C" sectional expander.

Next clean the fire-box ends of the flues and place them in the flue holes in the back flue sheet, fitting them neatly in the copper ferrules.

Flues shall project through the back flue sheet 3/16 inch and through the front flue sheet not less than ⅛ inch nor more than ¼ inch.

The preliminary setting and gaging shall be done with form F straight gaging expander.

Flues shall be tightened and flared in the back flue sheet with Faessler roller expanders, having flaring rolls.

Expanding flues in the back flue sheet shall be done with a form B modified sectional expander. With a No. 60 pneumatic hammer, drive the expander pin until it is fairly solid. Draw out the pin and give the expander a turn equal to one-half a section. Drive the pin a second time. Withdraw pin and turn the expander a distance equal to one quarter of a section. Drive the pin a third time and this will give the full set of the expander.

After expanding, examine flues carefully. Those which have split in the recess formed by the expander, and flues that have opened from the end into the sheet, shall be removed and replaced before beading.

The firebox end of all flues shall be beaded. This shall be done with a standard beading tool and a short stroke pneumatic hammer. Care must be taken to see that nothing enters between the bead and flue sheet. Hold the beading tool so as to give the bead an outward set and get it down to the sheet without raising a burr on the inside or marking the outside of the sheet with the heel of the beading tool. The center line of the beading tool shall always be inside the line of the flue. With a small chisel and hand hammer, remove any burrs that may be raised.

Two-inch and 2¼-inch flues on Classes A, C, G, K, T, and U shall not be welded except those behind the brick arch. Superheater flues on Classes A, G and K and all flues on Classes F, H, I, J, L, R and Y shall be welded.